Scope, Related Work, and Positioning

Scope, Related Work, and Positioning I

This section situates the exemplar and states explicit boundaries. The goal is not to compete with comprehensive treatments of exploratory data analysis [@tukey1977eda] or statistical graphics [@wilkinson2005grammar], but to show how a minimal, test-backed EDA story fits the template's reproducibility and rendering stack [@peng2011reproducible].

Exploratory data analysis I

The practice of examining a dataset before formal modelling — looking at distributions, missingness, and relationships — traces to Tukey's foundational work [@tukey1977eda] and is supported in modern practice by the pandas data analysis toolkit [@mckinney2010pandas] and the broader scientific-Python stack [@harris2020numpy]. The present manuscript restricts attention to a **first pass** on a **small tabular dataset**: load, surface missingness, compute descriptive statistics and per-group means, and rank features by Pearson correlation.

Modelling and inference (out of scope) I

What comes *after* a first EDA pass — hypothesis testing, regression, dimensionality reduction, or predictive modelling — is deliberately **out of scope**. The exemplar keeps the analysis minimal so the architectural lesson (notebook -> tested src extraction) stays visible rather than buried under statistical machinery.

What this project proves about the template I

The analytical steps here are standard. The **non-standard** contribution is procedural: the same tested functions in `src/eda/` back the interactive notebook, the headless analysis script, and this manuscript, so the figures and the summary table always refer to the same code. That pattern is what downstream projects should copy — whether the domain is survey data, sensor logs, or experimental measurements.

Explicit limitations I

1. **Dataset size**: a single small synthetic cohort (120 rows) is used for transparent, fast reproduction; no large-scale or streaming data is handled.
2. **Cleaning policy**: only listwise deletion is implemented; imputation is left as a documented extension.
3. **Correlation method**: only Pearson correlation is computed; rank-based (Spearman) or non-linear association measures are out of scope.
4. **Statics only**: the library returns plot-ready data; interactive widgets and dashboards are not part of the exemplar.

These limitations are intentional: they narrow the surface so that the reproducibility concerns — tested functions, a thin script, and a structurally verified notebook — remain visible rather than buried under analytical complexity.